

EMEM: A Content-Addressed, Verifiable Earth-Memory Protocol for AI Agents over Foundation-Model Embeddings

Jaya Kumari · Avijeet Singh — Vortex AI · doi:10.5281/zenodo.20706893 · claims pinned to emem 64cfae5; every live value below obtained from emem.dev on 29 Sep 2026 — integrity checks (A1–A3, A8) re-run with independent code

Two agents working on the same place have no way to know they are working on the same observation.

Each carries its own description, in its own words, inside its own context. Descriptions drift, and nothing in either model can detect it.

emem moves that identity out of the model into a shared record. **One place → one address. One observation → one signed fact.** An agent hands another party a token; the token dereferences to exact bytes; the receiver checks those bytes without trusting the sender, the network, or the server that served them.

The one rule: the bytes hash to the id, and the signature checks against a published key. Everything else is convenience.

Earth observation is the first and largest corpus the protocol serves, not its definition.

A1 A self-certifying observation you can rebuild

A fact is (**cell**, **band**, **tslot**, **value**, **uncertainty**, **sources**, **derivation**, **signer**, **signed_at**), encoded as deterministic CBOR (declaration order, NaN/−0 canonical) and named by BLAKE3 of those bytes. The derivation carries enough to recompute the value offline.

```
LIVE · VERIFY_NDVI.PY
fact oj5cecci... 1,115 bytes   re-hash MATCH
scene S2A_MSIL2A_20260925T054251_R005_T43SFS
DN B08 3502 B04 1900 B0A offset −1000 (PC, PB ≥ 04.00)
signed 0.4708994708994709 0x3fde233788cde233
rebuilt 0.4708994708994709 0x3fde233788cde233 bit-identical
same cid under another cell → HTTP 409
```

PRIOR ART STAC file:checksum / Merkle hash assets; RDA Dynamic Data Citation cites a stored query + timestamp.
NEW to our knowledge, an identifier for the individual value, carrying its own recomputation.
LIMIT signer and signed_at are hashed: a CID names one attestation, not the observation. Scenes are pinned by id, not by hash.

A3 “Looked, nothing there” is signed; “could not look” is not

An upstream’s definitive no-data becomes an **Absence** fact — signed, citable, its reason bound by hash. Timeouts and upstream errors return an **unsigned** skip note instead, so the two stay distinct.

```
LIVE · VERIFY_ABSENCE.PY · (−12.26, −140.15), OPEN OCEAN
kind absence value null
reason "Cop-DEM GLO-30 publishes no tile ... returned 404"
reason_cid yjtj52cf... re-derived locally ✓ receipt valid
(−82.5, 45.0) indices.ndvi → skip, upstream_error, unsigned
```

PRIOR ART DNSSEC NSEC/NSEC3; sparse-Merkle non-inclusion proofs; authenticated spatial range queries.
NEW to our knowledge, a signed EO distinction between absence and failure to observe.
LIMIT a signed assertion, not authenticated denial against a committed set; the reason hashes prose, not the upstream response.

A5 A claimed derivative is recomputed, and the gap reported in ULPs

/v1/derive resolves every parent token (mislabelled parent → 409, missing → 404), re-runs the op, and promotes the result to **deterministic_index** only on a match: exact for leaves, **≤ 4 ULP** for mean/sum over > 2 parents. It always returns the rule and the measured gap.

```
LIVE · SIGNED WITH A THROWAWAY KEY
exact delta      ulp_gap 0      verified
sum, +2 ULP      ulp_gap 2 (tol 4) verified
sum, +9 ULP      ulp_gap 9 (tol 4) refused → model_output
uncited “35 °C in Bengaluru on 2026-09-28”, gating on deny CLAIM_UNGROUNDED
```

PRIOR ART Nix / Bazel content-addressed derivations; bitwise vs ensemble-statistical reproducibility in Earth-system models.
NEW to our knowledge, hermeticity classes with a declared, measured ULP bound returned with each EO verdict.
LIMIT tier 1 covers scalar delta/mean/sum and registry expressions; arbitrary code and multi-hop lineage are not re-run.

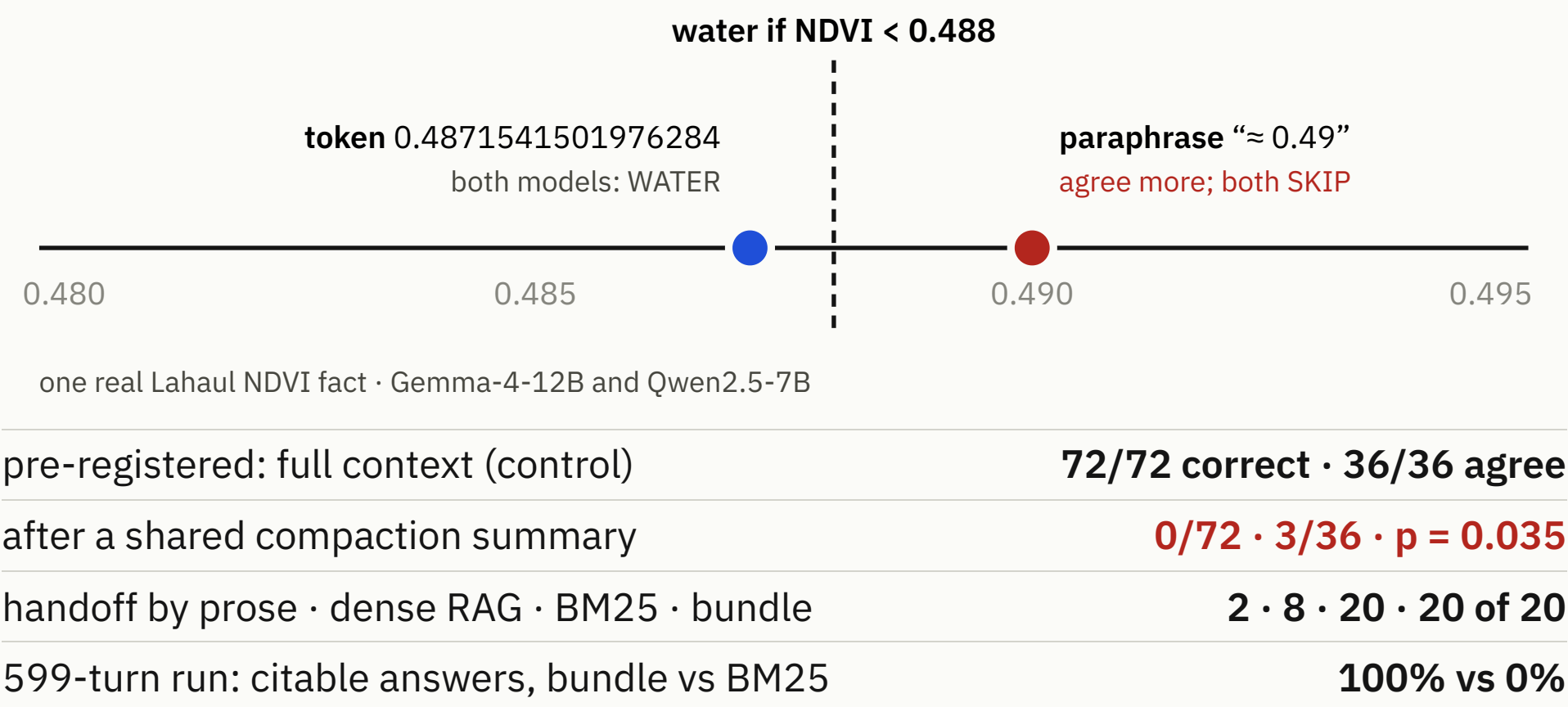
A7 An agent’s prose is checked against the values it cites

guard/verdict reads a draft, resolves each cited token, and compares the written number with the signed value at the writer’s own precision. Refusals are typed and carry the fix; a refused write returns the exact digest to sign.

```
LIVE · CITED FACT = 28.0 °C (MET.NO, BENGALURU)
“... was 35 °C (token)”      deny PROV_VALUE
“... was 28.0 °C (token)”    allow
same cid cited under another cell deny PROV_BYTES
receipt “35 °C in Bengaluru on 2026-09-28”, gating on deny CLAIM_UNGROUNDED
```

PRIOR ART signed receipts for MCP tool calls (2025–26); output guardrails, incl. GeoGuard at this workshop.
NEW to our knowledge, value-level checking of prose against signed EO facts, with citation strength typed.
LIMIT advisory on emem.dev; claim gating off by default; band-to-sentence matching not checked.

B1 · Agreement between agents is not evidence of correctness



Fisher one-sided; 0/72 ⇒ 95% upper bound ≈ 4.1%; one host, open models, no independent replication. BM25 ties on short handoffs. The phenomenon is known — Perez et al. ICLR 2025 (telephone game); Laban et al. 2505.06120; “The Deliberative Illusion” 2606.03032; “Lost in Compaction” 2608.11242; ARC 2607.25066 (pointer-based compaction). The contribution is a mechanism that detects drift of cited values.

B2 · A signed record of agents correcting each other — and us

69 attesters, 37,020 notes, 4,311 addressed to a peer, on at least five model families. A disagreement between two agents is itself a durable, addressable object a third party can fetch and check. 19 claims have been withdrawn — each a signed record beside the claim it retracts.	
per-encoder change calibration — shown impossible by an outside agent ($\partial^2 S / \partial c \partial x = -\partial \log L / \partial x$ vanishes on flat fields)	withdrawn
headline 1.000 counted empty answers → 0.989	corrected
cell64 cheaper for LLMs than H3 — 12.5 vs 8.5 tokens	refuted
handing a model the opaque id — tokenizers cut it identically 0–4% of the time; the descriptor lat, lng@date@band 100%	replaced
emem vs peer agent-memory products	not tested

Record: whitepaper v3 §2–3; counts from the live responder. geo.qa and other Vortex deployments are not independent adoption. Verifiability is not accuracy; a signature says who wrote a thing, never that it is true.

TOKEN FAMILY — OPENLY UNEQUAL		
TOKEN	DEREFERENCES TO	A MATCH PROVES
emem:fact:	exact stored bytes (32-byte BLAKE3)	BYTES
emem:raster:	signed derivation of an f32 grid	BYTES via artifact
emem:cube:	signed manifest of rasters in time	BYTES via members
emem:bundle:	a set of facts (≤ 256)	SET · members bytes
emem:entity:	a registered identity (16 B)	SHARED NAME
emem:cell:	nothing — refused by design	ADDRESS ONLY

A guard that checks citations treats only the first four as evidence. Treating an entity as a content commitment is the likeliest misuse of the protocol.

A2 Two clocks: when the ground was so, and when the system knew it

tslot is valid time (days since 1970 for fast bands; 0 for static). signed_at is transaction time. as_of_signed_at replays a key’s full signed history.

```
LIVE · VERIFY_BITEMPORAL.PY · BENGALURU · COPDEM30M.ELEVATION_MEAN
as known on 2026-05-01 no fact
as known on 2026-06-15 918.0 m open_meteo (Cop-DEM 90 m) re-hash ✓
as known on 2026-08-12 915.07 m Cop-DEM GLO-30 COG re-hash ✓
as known on 2026-09-29 915.07 m signed 2026-09-28 re-hash ✓
the upstream changed provider; the May answer still resolves and verifies
```

PRIOR ART SQL:2011 bitemporal periods; XTDB / Datomic as-of; Zep bitemporal agent memory (arXiv 2501.13956).
NEW to our knowledge, signed transaction time on EO values served to agents; a later agent can replay what an earlier one was told.
LIMIT without an explicit tslot the same query timed out (504) on this 280-fact cell.

A4 Disagreement is kept and returned, not fused

Every attestation at a (cell, band, tslot) key is retained. memory_contradictions returns disagreement per key — scalar spread, 1 – mean cosine for vectors, 1 – mode share for classes — and flags a signer that switched provider or recipe.

```
LIVE · BENGALURU ELEVATION KEY
disagreement_scope same_attester_provider_substitution
providers open_meteo_copdem90m01 → copernicus_dem_30m_aws_pixel01
attestations 8 (918.0 x1, 915.07 x7)
```

PRIOR ART truth discovery (Li et al. 2016) resolves conflicts; land-cover disagreement maps (Fritz et al. 2011); Dempster–Shafer conflict mass — usually in order to fuse.
NEW a design stance, not an algorithm: disagreement is a query answer with provenance.
LIMIT one responder today, so no genuine multi-party data; the severity scale is uncalibrated for bands without a registry range.

A6 Fields and fields-through-time as signed derivations

A raster is a canonical f32 grid whose hash sits in a signed record (AOI, band, scene, cloud, angles, anchors). A cube is a signed manifest of rasters that states, per member, **how far the scene is from the date asked for**.

LIVE · BAND_CUBE · S2.B08
asked scene used days away
15 May 10 Jun 26
15 Jun 30 Jun 15
15 Jul 03 Aug 19

PRIOR ART openEO / Earth Engine datacubes; STAC items.
NEW a content-addressed, signed cube manifest that records its temporal substitutions.
LIMIT scene choice uses scene-level cloud cover; a re-mint yields a new token for identical pixels (signed_at is hashed).

A8 An append-only record anyone can audit offline

RFC 6962 Merkle log over BLAKE3 with signed tree heads, inclusion and consistency proofs. Receipt v2 signs the Merkle proof **or an explicit ABSENT marker**, so stripping a proof or downgrading the version breaks the signature.

```
LIVE · VERIFY_LOG.PY, VERIFY_RECEIPT_TAMPER.PY (WRITTEN FROM RFC 9162)
STH signature valid tree size 2,538,190
inclusion of leaf 1,234,567 true 1-bit flipped leaf false
consistency 1,000,000 → 2,538,190 true
receipt as served valid · proof stripped invalid · relabelled v1 invalid
```

PRIOR ART Certificate Transparency, Trillian, Sigstore Rekor, Go sumdb witnesses.
NEW nothing cryptographic — a careful application to EO facts consumed by agents.
LIMIT the co-signing witness (geo.qa) is a second node on its own key, run by the same organisation; read federation does not exist.

Reproduce

```
# pip install blake3 cbor2 pynacl
python verify_fact.py # bytes → id
python verify_ndvi.py # A1
python verify_bitemporal.py # A2
python verify_absence.py # A3
python verify_log.py # A8
python verify_receipt_tamper.py \
  receipt_2026-09-29.json

# agents: MCP (18 core tools)
claude mcp add --transport http \
  emem https://emem.dev/mcp
# Python: pip install ememdev
# node: docker run -p 5051:5051 \
# ghcr.io/vortex-ai/emem:latest
```

Scripts
every live value here

Source
Apache-2.0

Paper
Zenodo

Not claimed. Anything running in orbit (edge signing is simulated; 0.17% of a 40 ms frame). Hardware-attested devices. Read federation. Formal proofs or full conformance vectors. Triple-encoder change consensus (retired). The accuracy of any observation.